Psychometric Project Instructions

Overview

The goal of this project is for you to design your own questionnaire. If you're taking this course, you're probably a psych major, and if you're a psych major, there's a decent chance you'll be tasked with using *some* kind of psychological instrument at some point in your career. Even if you never have to design an instrument yourself, building one will help understand the various problems that are inherent to measurement. Hopefully it also helps illustrate some of the key topics we've been covering this semester.

Selecting your topic:

You will first need to pick a topic for your scale to measure. I would suggest picking something kind of narrow, but not so narrow that you can't break into subcategories. Psychological disorders are a common starting point (e.g., depression, anxiety, etc.), as well as moods (e.g., happiness, sadness, anger), and traits (extraversion, openness, need for belonging). I would encourage you to pick something that is relevant to your own interests. If you aren't sure if a topic is suitable or if you are struggling to come up with something, do yourself a favor and come talk to me!

Whatever you end up picking, you will need to identify **three** subcomponents of whatever you are measuring. For example, you might hypothesize that depression is based on feelings of worthlessness, suicidal tendencies, and lack of energy/motivation. For each of these subcomponents, you'll need to come up with at least **Seven Items** that you think measure these constructs. (Feel free to come up with more).

Some things to keep in mind before getting started:

A good questionnaire has three components:

- Discrimination
- Reliability
- Validity

Discrimination: This component deals with item selection. Essentially, people who have different levels of a particular construct should give different answers to the items you select. Essentially:

- People with the same levels of a construct should answer the same way
- People with different levels should answer differently
- The degree of difference between levels should approximate the degree of difference between scores

Validity: This deals with whether your items actually measure what you have designed them to measure. Do scores on an anxiety scale actually measure anxiety, do depression scores actually reflect depression, etc.

When selecting your items, its important for you to consider *content validity* or how well the individual items match with the construct you are measuring

Factorial validity is also important. As part of your analyses, you will conduct a factor analysis on your data to ensure that items are assessing what they intend to. Because you will have three subscales, your factor analysis should reflect three sets of items.

Reliability: Finally, your questionnaire should be reliable. Basically, can your questionnaire produce the same results under the same conditions? For example, a valid IQ test should produce the same results each time the same person takes it.

You will need to assess your reliability by computing the split-half reliability for each subscale (Cronbach's alpha)

Designing your Questionnaire:

Step 1 – **Choose a Construct:** Okay, this is where the fun starts! You need to start by deciding what you want to measure. I would suggest hopping on google scholar and taking a look at what has already been done. I don't expect your search to be comprehensive, but I do expect you to get some basic background information on your construct, why its important to measure it, and how it might relate to other psychologically important things. For example, if you are interested in Empathy, you might look at this in terms of Client-Centered therapy (having empathy might be important for a therapist taking this approach). Thus, having a scale to measure empathy would be important for training therapists to use this approach. Basically, you should be able to justify why this is important, not just because you want to do it.

S**tep 2 – Pick your scale:** A fundamental issue in scale development is how you want respondents to answer questions.

Examples: Binary responses ("yes-no"), Likert scales (5-point; Strongly disagree → Strongly agree)

Whatever your items are, they should encourage participants to use all points of the scale. Your scale should also be consistent for all items (don't use a 5-point scale for some items, a binary scale for others, etc.)

Step 3 – Generate your Items: Once you've figured what construct you want to investigate and the type of scale you plan to use, now you need write some items. Remember, you'll need to come up with at least 21 items total (7 for each subscale). The best way to generate items is to brainstorm. I would encourage you to ask a small sample of people (friends, family, etc.) to list as many facets of your construct as they can think of. Look for specific themes in their answers. For example, if you are interested in exam anxiety, you might ask 5 of your friends to list out as many things as possible about exams that make them anxious. Afterwards, rephrase your sample's suggestions into items that fit with

the scale that you selected. I would encourage you to come up with about twice as many items as you actually need, and then eliminate items until you have 21 that you are comfortable with. If you want to include more than 21 items, that's fine! But don't include more than 30!

As you work on writing your items, think about how the questions are worded. Wording can potentially bias participants toward certain responses, so be mindful of this.

Also, be cautious of response bias. I would encourage you to reverse score a few items to keep participants from always giving the same answer (e.g., marking on 5's on teaching evals). **If you decide to reverse score, be sure to invert these items BEFORE running your analyses!**

Step 4 – Compile your items in a word document: The last step is to build the actual instrument. Use the template on D2L to build your measure. Don't put your name on the actual instrument – put it on the title page instead. Please print this document and bring it with you to class on 10/3.